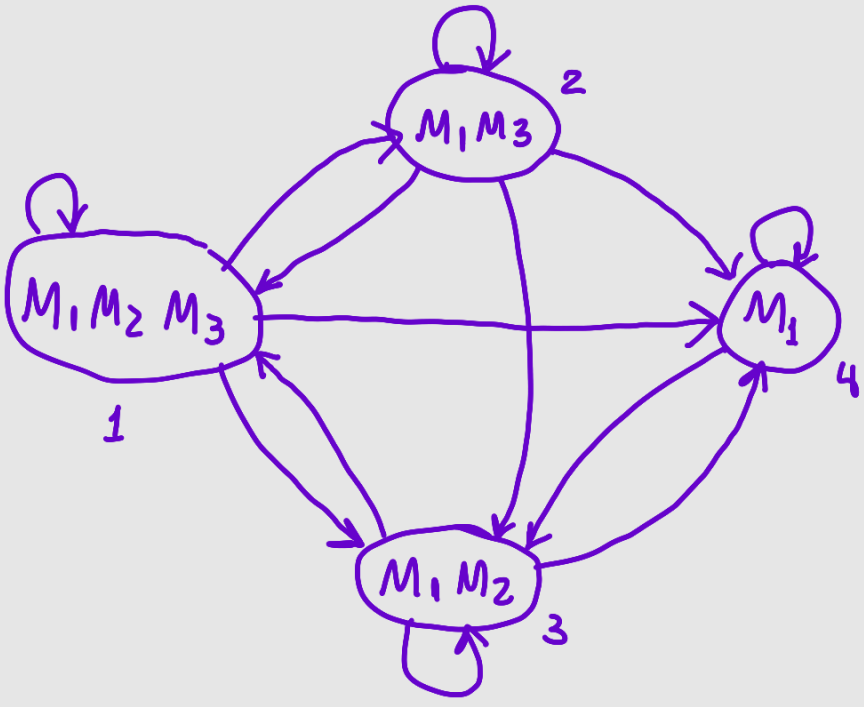
② $p_1 > p_2 > p_3$

Estado: máq. funcionando



$$P_{14} = (1-p_1) \cdot (1-p_2) \cdot (1-p_3)$$

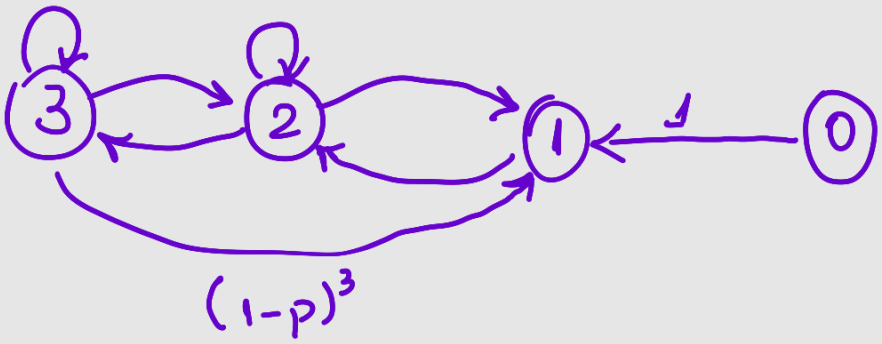
Matriz: 4x4

$$X_1 = X_0 \cdot P = [0{,}788 \; - \; - \; -]$$

Condição somente uma distribuição estacionária

<B 1 classe fechada

(1)



$$P = \begin{array}{c} \\ 3 \\ 2 \\ 1 \\ 0 \end{array} \begin{array}{c} \begin{array}{cccc} 3 & 2 & 1 & 0 \end{array} \\ \begin{bmatrix} 0,896 & 0,096 & 0,008 & 0 \\ 0,64 & 0,32 & 0,04 & 0 \\ 0 & 0,8 & 0,2 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

$X_0 = [1/4 \quad 1/4 \quad 1/4 \quad 1/4]$

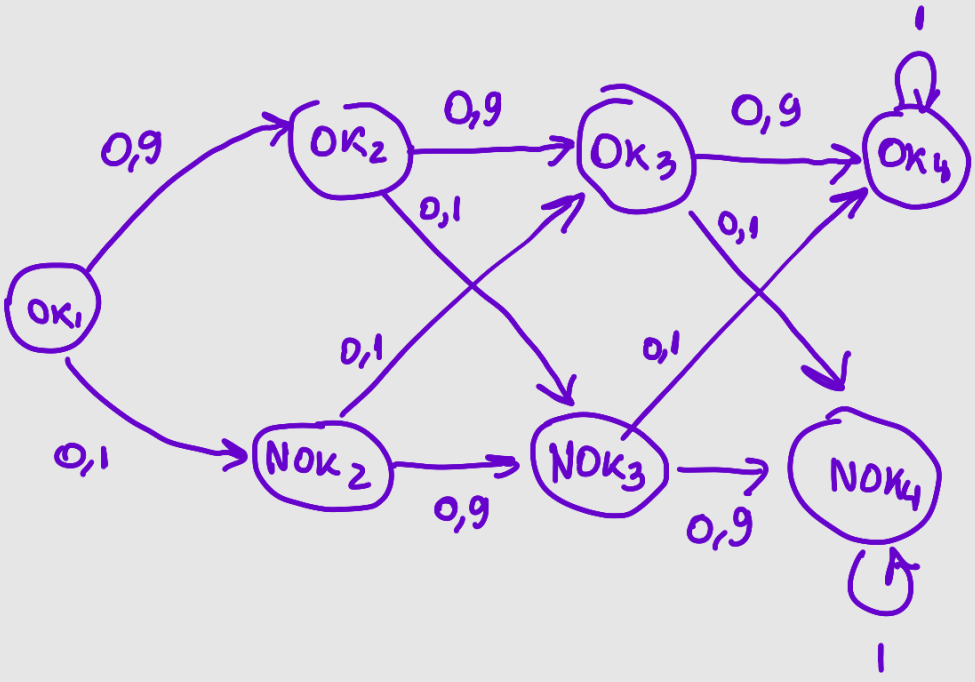
$X_1 = X_0 \cdot P$

$X_2 = X_1 \cdot P$

$X_2 = [0,538 _ _ _]$

$p_3 = 0,84$

(3)



$X_0 = [1\ 0\ 0\ 0\ 0\ 0\ 0\ 0]$

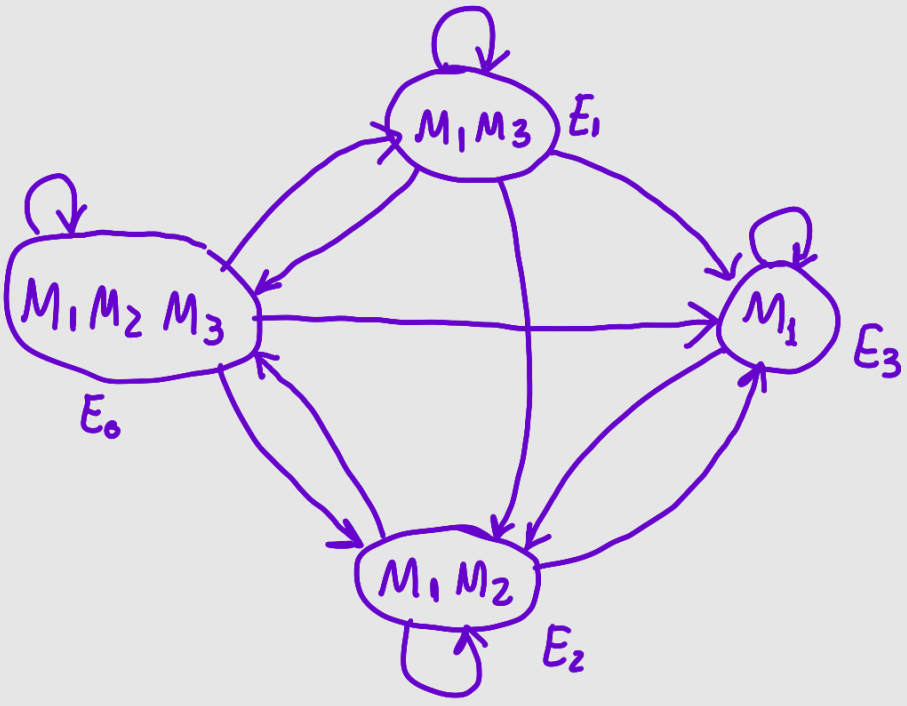
$X_1 =$

$X_2 =$

$X_3 = [0\ 0\ 0\ 0\ 0\ \ 0,756\ \ 0,244]$

$p_1 = 0,8 \qquad p_2 = 0,7 \qquad p_3 = 0,6$

(2)



$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} p_{00} & p_{01} & p_{02} & p_{03} \\ p_{10} & p_{11} & p_{12} & p_{13} \\ p_{20} & 0 & p_{22} & p_{23} \\ 0 & 0 & p_{32} & p_{33} \end{bmatrix} \end{array} \begin{array}{c} \\ 0,336 \\ 0,084 \\ 0,144 \\ 0,224 \end{array}$$

$$p_{00} = (p_1 \cdot p_2 \cdot p_3) + (1-p_1) \cdot p_2 \cdot p_3 + p_1 \cdot (1-p_2) \cdot p_3 + p_1 \cdot p_2 \cdot (1-p_3) = 0,788$$

$$p_{01} = (1-p_1) \cdot (1-p_2) \cdot p_3 = 0,036$$

$$p_{02} = (1-p_1) \cdot p_2 \cdot (1-p_3) + p_1 \cdot (1-p_2) \cdot (1-p_3) = 0,152$$

$$p_{03} = (1-p_1) \cdot (1-p_2) \cdot (1-p_3) = 0,024$$

$$p_{10} = p_1 \cdot 1 \cdot p_3 = 0,480$$

$$p_{11} = p_1 \cdot (1-p_2) \cdot p_3 = 0,144$$

$$p_{12} = p_1 \cdot 1 \cdot (1-p_3) = 0,320$$

$p_{13} = (1-p_1)\cdot(1-p_3) = 0{,}080$

$p_{20} = p_1 \cdot p_2 \cdot p_3$

$p_{21} = 0$

$p_{22} = p_1 \cdot p_2 \cdot (1-p_3)$

$p_{23} = (1-p_1)\cdot(1-p_2)$

$p_{30} = 0$

$p_{31} = 0$

$p_{32} = p_1$

$p_{33} = p_1 \cdot (1-p_2)\cdot(1-p_3)$